

Admin Backend — Comprehensive Wireframe Analysis

Source: 12 admin wireframe HTML files

Date: June 5, 2026

Purpose: Complete feature, data-model, flow, and integration analysis prior to implementation

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1. Executive Summary

The admin backend is a **separate, internal-only console** (not accessible to front-end collectors) that provides Luciteria staff with platform management capabilities across **six functional domains**:

Domain	Screens	Purpose
Authentication	Login	Standalone email/password auth (not Shopify)
User Management	Users, Freeze Modal, Award Milestone, Bulk Notification	View, search, freeze/unfreeze, award milestones, bulk-notify collectors
Admin Management	Manage Admins, Add Admin Modal	CRUD admin accounts with primary/non-primary roles
Analytics	Analytics Dashboard	Platform-wide user, collection, format, and source metrics
Catalog Management	Formats, Set Builder, SKU Sync	Manage formats, build curated sets, sync cross-platform pricing
Demand Intelligence	Demand Signals	Restock prioritization from wishlist/watchlist/waitlist data

Two distinct sidebar navigation schemas appear across the wireframes (see [Section 8](#)), suggesting an iteration occurred during design.

2. Screen-by-Screen Analysis

2.1 Admin Login (`admin-login.html`)

Purpose: Standalone authentication gate for the admin console.

Features:

- Email + password login form
- “Remember me” checkbox
- “Forgot password?” link
- Password visibility toggle (eye icon)
- Info banner: “This is the admin console. User login is via Shopify.”

UI Components:

- Card-based centered layout (420px)
- Brand header with cube icon + “Luciteria / Collector Cabinet”
- Two text inputs (email, password) with left-side icons
- Single CTA button: “Sign In” (gray-700 bg)

Data Touched:

- `AdminUser.email` , `AdminUser.passwordHash`
- Session/token creation

2.2 User Management (admin-users.html)

Purpose: Browse, search, filter, and act on all collector accounts.

Features:

- **Summary stats:** “247 total users · 189 active”
- **Search:** By name or email (text input)
- **Filters:** Motivation (All / Inventory / Social / Acquisition / Investment / Exploring), Tier (All / Free / Lucite Pro / Curator), Status (All / Active / Frozen)
- **Bulk selection bar:** Shows “3 users selected” with bulk actions: “Send Notification”, “Freeze Accounts”
- **User table columns:** Checkbox, Name, Email, Owned count, Wanted count, Status badge, Last Login, Actions (ellipsis menu)
- **Per-row actions (active user):** Freeze Account, Award Milestone, View Details
- **Per-row actions (frozen user):** Unfreeze Account, View Details
- **Pagination:** Prev / Next with “Showing 8 of 247 users”
- **Export Data** button (top-right)

UI Components:

- Sidebar navigation (Schema A — see Section 8)
- Search + filter toolbar (white card)
- Bulk action bar (gray-200 bg, conditional on selection)
- Data table (grid layout, 8 columns)
- Row-level hover dropdown menus
- Pagination controls

Data Touched:

- `User.*` (name, email, ownedCount, wantedCount, status, lastLogin, motivation, tier)
-

2.3 Freeze Account Modal (admin-users-freeze-modal.html)

Purpose: Confirm freezing a specific user’s account with a required reason.

Features:

- Shows target user (avatar initials, name, email)
- **Required** reason textarea
- Info note: “User will have read-only access. They can still view their collection but cannot make changes.”
- Cancel / “Freeze Account” buttons

UI Components:

- Modal overlay (480px, dimmed background)
- User identity card (gray-50 bg)
- Textarea (required)
- Info banner
- Two-button footer (Cancel + Confirm)

Data Touched:

- `User.status` → “frozen”
 - `FreezeLog.reason` , `FreezeLog.frozenAt` , `FreezeLog.frozenBy`
-

2.4 Admin Users Management (`admin-manage-admins.html`)

Purpose: List all admin users, manage activation status, add/remove admins.

Features:

- **Header:** "All admin users have full access to all features."
- **Table columns:** Name (with avatar initials), Email, Date Added, Status, Last Login, Actions
- **Primary admin row:** Labeled "You · Primary" — no action buttons, cannot self-remove/deactivate
- **Other admin actions:** Deactivate / Remove (text links)
- **Inactive admin actions:** Reactivate / Remove
- **" + Add Admin User "** button (top-right)
- **Security note:** "Admin users can access all data and perform all actions. Only grant admin access to trusted team members."
- **Footer:** "4 admin users · You cannot remove yourself or deactivate the primary admin."

UI Components:

- Data table (6 columns)
- Status badges: Active (filled circle), Inactive (pause icon)
- Security info banner (shield icon)

Data Touched:

- `AdminUser.*` (name, email, dateAdded, status, lastLogin, isPrimary)
-

2.5 Add Admin User Modal (`admin-add-admin-modal.html`)

Purpose: Create a new admin account.

Features:

- **Form fields:** First Name, Last Name, Email (with helper: "Must be a valid email address. Login instructions will be sent here.")
- Temporary password: Auto-generated, shown in readonly mono input, with Copy + Regenerate buttons
- Info note: "The new admin will receive an email with login instructions and will be prompted to change their password on first login."
- Checkbox:* "Send welcome email now" (checked by default)
- Cancel / "Add Admin User" buttons

UI Components:

- Modal overlay (520px, dimmed background)
- 2-column name fields
- Readonly password field with action buttons
- Info banner
- Checkbox
- Two-button footer

Data Touched:

- Creates `AdminUser` record (firstName, lastName, email, tempPassword, status="active")
 - Triggers welcome email (optional)
-

2.6 Analytics Dashboard (admin-analytics.html)

Purpose: Platform-wide metrics across users, collections, formats, and acquisition sources.

Features:

User Metrics Section

- **KPI cards (4):** Total Users, Active Users (30d), Inactive, Avg Logins/Week
- **User Activity table:** Name, Last Login, Total Logins, Motivation — with segment-by-motivation dropdown filter

Collection Metrics Section

- **Per User Average:** Owned / Wanted / Wishlist counts
- **Platform Wide Total:** Owned / Wanted / Wishlist aggregate counts

Product Types Section

- **Format Breakdown bar chart:** 10mm, 25.4mm, 50mm, Lucite, Ampoule — horizontal bars with counts
- **Top 10/20 Most Collected Elements** (expandable/collapsible list, ranked by owner count)
- **Top 10/20 Most Wanted Elements** (expandable/collapsible list, ranked by wishlist demand)

External Sources Section

- **Source ranking table:** 18 sources listed (eBay, Amazon, Luciteria Direct, RotoMetals, Element Sales, Etsy, Alibaba, Metallium, Luciteria Gift, Mineralogical Research, Sigma-Aldrich, Edmund Optics, Smart Elements, Goodfellow, Local Mineral Show, Estate Sale, University Surplus, Other/Private)
- Collapsible: “Show Less (Top 6)” / “Show More”

Time Range Filter: Global selector — Last 7 days / Last 30 days / Last 90 days / All time

UI Components:

- 4-column KPI card grid
- Data table with segment filter
- 2-column metric cards with sub-grids
- Horizontal bar chart (CSS-based)
- Two side-by-side ranked lists (expandable)
- Source count table (collapsible)

Data Touched:

- Aggregated from `User`, `ElementSample`, `WishlistItem`, `WatchlistItem` tables
- `ElementSample.source` field (for external sources)
- `ElementSample.format` (for format breakdown)
- `User.motivation`, `User.lastLogin`, `User.totalLogins`

2.7 Collection Formats (admin-formats.html)

Purpose: CRUD management of the physical formats elements can be collected in.

Features:

- **Table columns:** Format Name (with icon), Used By (user count), Active Status, Actions
- **Listed formats:** 10mm Cube, 25.4mm Cube, 50mm Cube, Lucite Cube, Ampoule, Other
- **Inline editing:** One row shown in edit mode with text input + “Editing...” label + Save/Cancel but-

tons

- **Row actions:** Edit (pen icon), Delete (trash icon)
- **Status badges:** Active (filled circle) or Inactive (hollow circle)
- **" + Add Format"** button (top-right)
- **Footer note:** "Format changes automatically apply to wishlist, collection tracking, and progress views."

UI Components:

- Data table (4 columns)
- Inline edit mode (replaces static text with input)
- Icon-based format identifiers (cube, gem, vial, ellipsis)
- Status toggle badges

Data Touched:

- `Format.id` , `Format.name` , `Format.icon` , `Format.isActive` , `Format.usedByCount` (computed)

2.8 Collection Set Builder (`admin-set-builder.html`)

Purpose: Create curated sets of elements that can be sold as packages.

Features:

- **Breadcrumb:** "← Back to Sets" (implies a Sets list page exists but no wireframe provided)
- **Form fields:**
 - Set Name (text input)
 - Description (textarea)
 - Price (text input, formatted as "\$189.00")
 - Stock Allocation (text input, e.g. "25 units")
- **Tier Eligibility:** Checkboxes for Free / Collector / Curator
- **Element Selection:** Interactive periodic table grid — click to toggle elements; selected elements shown in dark (gray-700); legend: "Selected" vs "Available"; counter: "8 selected · click to toggle"
- **Live Preview Panel (sticky sidebar):**
 - Cover image placeholder (dashed border)
 - Set name, description
 - Summary: Elements count, Price, Stock, Tiers
 - Included elements list (symbol chips)
- **Action buttons:** "Save Draft" + "Publish Set"

UI Components:

- 3-column layout (2 form + 1 preview)
- Multi-section card form
- Interactive periodic table mini-grid (18-column CSS grid)
- Sticky preview sidebar
- Checkbox group (tier selection)
- Dual action buttons (draft/publish)

Data Touched:

- `CollectionSet.id` , `name` , `description` , `price` , `stockAllocation` , `status` (draft/published)
- `CollectionSetTier` (junction: set ↔ tier)
- `CollectionSetElement` (junction: set ↔ element)
- `CollectionSet.coverImage` (implied by placeholder)

2.9 SKU Sync Validation (`admin-sku-sync.html`)

Purpose: Cross-platform price and stock comparison between Shopify and eBay.

Features:

- **Header actions:** “Re-sync” button, “Export CSV” button
- **Last sync timestamp:** “Last synced 12 minutes ago”
- **Filters:** Status, Platform, Element (dropdowns) + SKU search input
- **Bulk toolbar:** Checkbox select-all, “3 selected”, “Update Price”, “Flag for Review” buttons
- **Data table columns:** Checkbox, SKU (monospace), Element, Shopify Price, eBay Price, Delta, Stock (Shopify), Qty (eBay), Status
- **Status badges:** OK, Too Low, Too High, Missing Data (dashed border)
- **Price delta:** Shown as signed dollar amount (\$0.00, -\$15.00, +\$34.00) — bold for non-zero
- **Missing data handling:** Em-dashes for unavailable eBay data
- **Pagination:** Prev/Next, “Showing 8 of 142 SKUs”

UI Components:

- Dual action buttons (top-right)
- 4-filter toolbar + search
- Bulk action bar
- Full HTML `<table>` (not CSS grid — different from other screens)
- Status badge variants (4 states)
- Pagination

Data Touched:

- `SkuMapping.sku`, `element`, `shopifyPrice`, `ebayPrice`, `priceDelta`, `shopifyStock`, `ebayQty`, `sync-Status`
- `SkuMapping.lastSyncedAt`
- Integration: Shopify Product API, eBay Listing API

2.10 Demand Signal Intelligence (`admin-demand-signal.html`)

Purpose: Restock intelligence powered by wishlist, watchlist, and waitlist activity.

Features:

- **KPI cards (3):**
- Total Waitlist Size: 1,284 (+12% vs last 30 days)
- Avg Priority Score: 68.4 (across 142 tracked SKUs)
- Pending Restocks: 17 (9 high priority)
- **Top Restock Priorities:** Ranked list of 6 items with priority scores (e.g., “Rhenium 10mm — 94”)
- **Most Wanted Elements bar chart:** By wishlist frequency — Rhenium (312), Hafnium (278), Iridium (241), Osmium (205), Gadolinium (188), Tantalum (164)
- **Waitlist Growth chart:** SVG line chart showing 8-week trend (W1-W8), upward trajectory

UI Components:

- 3-column KPI card grid
- 3-column content grid (list + bar chart + line chart)
- Numbered ranked list with score badges
- Horizontal bar chart (CSS-based)
- SVG line chart (hand-drawn polyline)

Data Touched:

- Aggregated from `WishlistItem` , `WatchlistItem` , `WaitlistEntry` tables
 - `DemandSignal.elementId` , `format` , `priorityScore` , `wishlistCount` , `waitlistCount`
 - Computed: priority score algorithm (undefined in wireframe)
-

2.11 Bulk Notification Modal (`admin-bulk-notification.html`)

Purpose: Send targeted or broadcast notifications to collectors.

Features:

- **Recipient indicator:** “Sending to 247 users” (dynamic based on segment)
- **Segment filter:** All Users / Inventory / Social / Acquisition / Investment / Exploring
- **Form fields:** Notification title (text input), Message (textarea)
- **Delivery method:** Radio buttons — “In-App only” (default) / “Include Email”
- **Live preview panel:** Shows notification card as it would appear in the user’s inbox (avatar, title, message, timestamp)
- Cancel / “Send Notification” buttons

UI Components:

- Modal overlay (760px, wide)
- 2-column layout inside modal (form + preview)
- Segment dropdown
- Radio button group
- Live preview card (white card on gray-50 bg)

Data Touched:

- Creates `Notification` records per targeted user
 - `Notification.title` , `message` , `deliveryMethod` , `sentAt` , `sentBy`
 - `Notification.segment` (for targeting logic)
-

2.12 Award Milestone Modal (`admin-award-milestone.html`)

Purpose: Manually award a collection milestone badge to a specific user.

Features:

- **Target user display:** Avatar initials, name, email (pre-populated from row context)
- **Milestone dropdown:** First 10 Elements, Reach 25 Elements, Reach 50 Elements, Noble Gases Complete, Transition Metals Complete, Full Lucite Format Set
- **Optional note:** Textarea for personal message
- **Checkbox:** “Notify user” (checked by default)
- Cancel / “Award Milestone” buttons

UI Components:

- Modal overlay (480px)
- User identity card
- Select dropdown (6 milestone options)
- Textarea (optional)
- Checkbox
- Two-button footer

Data Touched:

- Creates `UserMilestone` record (userId, milestoneId, awardedAt, awardedBy, note)
 - Optionally creates `Notification` for the user
-

3. Complete Feature List

Authentication & Access Control

#	Feature	Screen
1	Admin email/password login	Login
2	Remember me persistence	Login
3	Forgot password flow	Login
4	Password visibility toggle	Login
5	Session management	Implied

User Management

#	Feature	Screen
6	View all collectors (paginated table)	Users
7	Search users by name/email	Users
8	Filter by motivation	Users
9	Filter by subscription tier	Users
10	Filter by status (active/frozen)	Users
11	Bulk select users	Users
12	Bulk send notification	Users → Bulk Notification
13	Bulk freeze accounts	Users → Freeze Modal
14	Per-user: Freeze account (with reason)	Freeze Modal
15	Per-user: Unfreeze account	Users (row menu)
16	Per-user: Award milestone	Award Milestone
17	Per-user: View details	Users (row menu)
18	Export user data	Users

Admin Management

#	Feature	Screen
19	View all admin users	Manage Admins
20	Add new admin user	Add Admin Modal
21	Auto-generate temporary password	Add Admin Modal
22	Copy/regenerate password	Add Admin Modal
23	Send welcome email	Add Admin Modal
24	Deactivate admin	Manage Admins
25	Reactivate admin	Manage Admins
26	Remove admin	Manage Admins
27	Primary admin protection (cannot self-remove)	Manage Admins

Analytics

#	Feature	Screen
28	User KPIs (total, active, inactive, avg logins)	Analytics
29	User activity table (segmentable by motivation)	Analytics
30	Collection averages (owned/wanted/wishlist per user)	Analytics
31	Platform-wide collection totals	Analytics
32	Format breakdown bar chart	Analytics
33	Top collected elements (expandable top 10/20)	Analytics
34	Top wanted elements (expandable top 10/20)	Analytics
35	External source ranking (collapsible)	Analytics
36	Time range filter (7d / 30d / 90d / all time)	Analytics

Catalog Management

#	Feature	Screen
37	View all formats	Formats
38	Add new format	Formats
39	Inline rename format	Formats
40	Delete format	Formats
41	Toggle format active/inactive	Formats
42	Create collection set (name, description, price, stock)	Set Builder
43	Select elements for set via periodic table	Set Builder
44	Assign tier eligibility to set	Set Builder
45	Save set as draft	Set Builder
46	Publish set	Set Builder
47	Live preview of set	Set Builder
48	Set cover image upload (placeholder shown)	Set Builder

SKU Sync & Pricing

#	Feature	Screen
49	Cross-platform price comparison (Shopify vs eBay)	SKU Sync
50	Stock/quantity comparison	SKU Sync
51	Price delta calculation	SKU Sync
52	Status classification (OK / Too Low / Too High / Missing Data)	SKU Sync
53	Filter by status, platform, element	SKU Sync
54	Search by SKU	SKU Sync
55	Bulk update price	SKU Sync
56	Bulk flag for review	SKU Sync
57	Manual re-sync trigger	SKU Sync
58	Export CSV	SKU Sync

Demand Intelligence

#	Feature	Screen
59	Waitlist size KPI with trend	Demand Signals
60	Average priority score	Demand Signals
61	Pending restock count (with high-priority sub-count)	Demand Signals
62	Top restock priorities ranked list	Demand Signals
63	Most wanted elements bar chart	Demand Signals
64	Waitlist growth line chart (8-week trend)	Demand Signals

Notifications

#	Feature	Screen
65	Compose bulk notification (title + message)	Bulk Notification
66	Segment targeting (by motivation)	Bulk Notification
67	Delivery method selection (in-app / in-app + email)	Bulk Notification
68	Live notification preview	Bulk Notification
69	Per-user milestone notification	Award Milestone

4. Data Requirements

4.1 New Models Required

AdminUser

```

id          String    @id @default(uuid())
firstName   String
lastName    String
email       String    @unique
passwordHash String
isPrimary   Boolean    @default(false)
status      Enum      (active | inactive)
dateAdded   DateTime  @default(now())
lastLogin   DateTime?
mustChangePassword Boolean @default(true)

```

AdminSession

```

id          String    @id @default(uuid())
adminUserId String    → AdminUser
token       String    @unique
rememberMe  Boolean    @default(false)
expiresAt   DateTime
createdAt   DateTime  @default(now())

```

FreezeLog

```

id          String    @id @default(uuid())
userId      String    → User
frozenBy    String    → AdminUser
reason      String
action      Enum      (freeze | unfreeze)
createdAt   DateTime  @default(now())

```

Milestone

```

id          String    @id @default(uuid())
name        String    @unique
description  String?
threshold   Int?      (for count-based milestones)
type        Enum      (count | category_complete |
                      format_complete)
sortOrder   Int

```

UserMilestone

```

id          String    @id @default(uuid())
userId      String    → User
milestoneId String    → Milestone
awardedBy   String    → AdminUser
note        String?
awardedAt   DateTime  @default(now())
@@unique([userId, milestoneId])

```

```

Format
id          String    @id @default(uuid())
name        String    @unique
icon        String    (FontAwesome class reference)
isActive    Boolean    @default(true)
sortOrder   Int
createdAt   DateTime   @default(now())
updatedAt   DateTime   @updatedAt

CollectionSet
id          String    @id @default(uuid())
name        String
description  String?
price       Decimal
stockAllocation Int
coverImageUrl String?
status      Enum      (draft | published)
createdBy   String    → AdminUser
createdAt   DateTime   @default(now())
updatedAt   DateTime   @updatedAt

CollectionSetElement
id          String    @id @default(uuid())
setId       String    → CollectionSet
atomicNumber Int
@@unique([setId, atomicNumber])

CollectionSetTier
id          String    @id @default(uuid())
setId       String    → CollectionSet
tierName    String    (Free | Collector | Curator)
@@unique([setId, tierName])

SkuMapping
id          String    @id @default(uuid())
sku         String    @unique
elementName String
atomicNumber Int
format      String
shopifyPrice Decimal?
ebayPrice   Decimal?
priceDelta  Decimal? (computed)
shopifyStock Int?
ebayQty     Int?
syncStatus  Enum      (ok | too_low | too_high |
                        missing_data)
flaggedForReview Boolean @default(false)
lastSyncedAt DateTime?
createdAt   DateTime   @default(now())
updatedAt   DateTime   @updatedAt

```

```

DemandSignal (materialized / computed)

id          String  @id @default(uuid())
atomicNumber Int
format      String
wishlistCount Int
waitlistCount Int
priorityScore Float
restockPending Boolean @default(false)
computedAt  DateTime @default(now())

AdminNotification

id          String  @id @default(uuid())
title       String
message     String
segment     String? (null = all users)
deliveryMethod Enum (in_app | in_app_and_email)
recipientCount Int
sentBy      String  → AdminUser
sentAt      DateTime @default(now())

```

4.2 Modifications to Existing Models

Model	Change	Reason
User	Add <code>status</code> field (active/frozen)	Freeze/unfreeze functionality
User	Add <code>tier</code> field (free/lucite_pro/curator)	Filter and tier eligibility
User	Add <code>motivation</code> field (if not already present)	Filtering and segmentation
User	Add <code>totalLogins</code> counter	Analytics user activity table
User	Add <code>lastLogin</code> timestamp	User table + analytics
ElementSample	Ensure <code>source</code> field exists	External sources analytics
ElementSample	Ensure <code>format</code> references <code>Format</code> model	Format breakdown analytics
Notification (existing)	Add <code>adminNotificationId</code> FK	Link user notifications to admin broadcast

5. User Flows

5.1 Authentication Flow

Login Page → Enter Email + Password → [Valid?]
 → Yes → Redirect to Dashboard (or Users page)
 → No → Show error, stay on Login
 "Forgot Password?" → [Not wireframed → needs design]

5.2 Freeze/Unfreeze User Flow

Users Page → Select user row → Click ":" menu → "Freeze Account"
 → Freeze Modal opens → Enter reason (required) → Click "Freeze Account"
 → User status changes to "Frozen" → Modal closes → Table refreshes
 Users Page → Select frozen user → Click ":" menu → "Unfreeze Account"
 → [No modal shown – direct action? Or confirmation needed?]

5.3 Award Milestone Flow

Users Page → Select user row → Click ":" menu → "Award Milestone"
 → Award Modal opens → Select milestone from dropdown
 → Optionally add note → Toggle "Notify user"
 → Click "Award Milestone" → Record created → Notification sent (if checked)

5.4 Bulk Notification Flow

Users Page → Select multiple users (checkboxes) → Click "Send Notification"
 → Bulk Notification Modal opens → Select segment (optional refinement)
 → Enter title + message → Choose delivery method
 → Review preview → Click "Send Notification"
 → Notifications created for all targeted users → Modal closes

5.5 Admin Management Flow

Admin Users Page → Click "+ Add Admin User"
 → Add Admin Modal opens → Fill first name, last name, email
 → Copy auto-generated temporary password
 → Toggle "Send welcome email now" → Click "Add Admin User"
 → Admin record created → Welcome email sent → Modal closes → Table refreshes
 Admin Users Page → Click "Deactivate" on a row → [Direct toggle? Confirmation?]
 Admin Users Page → Click "Remove" on a row → [Confirmation modal? Not wireframed]

5.6 Format Management Flow

Formats Page → Click "+ Add Format" → [New row? Modal? Not wireframed]
 Formats Page → Click pen icon on row → Row enters inline edit mode
 → Modify name → Click "Save" (or "Cancel")
 Formats Page → Click trash icon → [Confirmation? What about "Used By" count?]

5.7 Set Builder Flow

[Sets List ☐ not wireframed] ☐ Click "Create Set" OR "Edit Set"

- ☐ Set Builder Page ☐ Fill metadata (name, description, price, stock)
- ☐ Select tier eligibility ☐ Click elements on periodic table
- ☐ Upload cover image ☐ Review live preview
- ☐ "Save Draft" or "Publish Set"

5.8 SKU Sync Flow

SKU Sync Page ☐ [Auto-loaded on page visit]

- ☐ Apply filters (status / platform / element / SKU search)
- ☐ Review price deltas and status flags
- ☐ Select rows ☐ "Update Price" or "Flag for Review"
- ☐ Click "Re-sync" to pull latest data
- ☐ Click "Export CSV" for offline analysis

6. UI Components Inventory

Shared / Global Components

Component	Used In	Notes
Admin Sidebar	All pages (post-login)	Two variant schemas (see Section 8)
Brand Header	Sidebar + Login	Cube icon + "Luciteria" / "Admin Console"
"Back to Collector view" link	Sidebar footer	Links back to collector-facing app

Data Display

Component	Used In	Notes
KPI Card	Analytics, Demand Signals	Large number + label + optional trend indicator
Data Table (CSS grid)	Users, Admins, Analytics, Formats	Consistent column header styling
Data Table (HTML table)	SKU Sync	Different pattern — uses <code><table></code> element
Ranked List	Analytics, Demand Signals	Numbered items with right-aligned values
Horizontal Bar Chart	Analytics, Demand Signals	CSS-based width percentages
SVG Line Chart	Demand Signals	Hand-drawn polyline in view-Box
Status Badge	Users, Admins, Formats, SKU Sync	Pill-shaped, variant by status
Pagination	Users, SKU Sync	Prev/Next + showing count

Forms & Inputs

Component	Used In	Notes
Text Input	Login, Add Admin, Set Builder, Bulk Notification	With and without icons
Textarea	Freeze Modal, Award Milestone, Set Builder, Bulk Notification	Various row counts
Select Dropdown	Users, Analytics, Award Milestone, SKU Sync, Bulk Notification	Custom styled with chevron
Checkbox	Users (bulk select), Add Admin, Set Builder (tiers), Award Milestone	Both standalone and in groups
Radio Button Group	Bulk Notification	Delivery method selection
Inline Edit Input	Formats	Replaces static text in table row
Password Input	Login, Add Admin	With visibility toggle / copy / regenerate

Modals & Overlays

Component	Used In	Dimensions
Freeze Account Modal	Users → Freeze	480px
Add Admin Modal	Admin Users → Add	520px
Bulk Notification Modal	Users → Notify	760px (2-column)
Award Milestone Modal	Users → Award	480px

All modals share: dimmed bg overlay, header with icon + title + close button, footer with Cancel + primary action button.

Action Elements

Component	Used In	Notes
Primary Button	All screens	gray-700/800 bg, white text
Secondary Button	All screens	white bg, gray border
Row Action Menu	Users	Hover-triggered dropdown
Inline Text Links	Admin Users	“Deactivate” / “Remove” / “Reactivate”
Icon Buttons	Formats	Pen (edit) / Trash (delete)
Bulk Action Bar	Users, SKU Sync	Conditional on selection

7. Integration Points

7.1 With Existing Collector Cabinet

Integration	Direction	Details
User data read	Admin ← Cabinet	Users table, ElementSample, WishlistItem, WatchlistItem read for admin views
User status write	Admin → Cabinet	Freeze/unfreeze changes User.status; cabinet must enforce read-only for frozen users
Notification delivery	Admin → Cabinet	AdminNotification → creates records in existing Notification table; appears in collector's notification inbox
Milestone awards	Admin → Cabinet	UserMilestone records must surface on collector's dashboard/profile
Format changes	Admin → Cabinet	Format CRUD propagates to collection page format selector, onboarding format cards, shop page format filter
Collection Set visibility	Admin → Cabinet	Published sets appear in collector-facing shop/collection areas

7.2 With Shopify

Integration	Direction	Details
Product price sync	Shopify → Admin	SKU Sync reads Shopify product prices via API
Stock levels	Shopify → Admin	SKU Sync reads inventory levels
Price updates	Admin → Shopify	“Update Price” action writes back to Shopify (implied)
Set publishing	Admin → Shopify	Published sets may need to create Shopify products/collections
User authentication	Shopify → Cabinet	Admin note confirms user auth is separate (“User login is via Shopify”)

7.3 With eBay

Integration	Direction	Details
Listing prices	eBay → Admin	SKU Sync reads eBay listing prices
Listing quantities	eBay → Admin	SKU Sync reads eBay available quantities
Price updates	Admin → eBay?	Unclear if “Update Price” also writes to eBay

7.4 Internal Computed Data

Computation	Source Tables	Output
User KPIs	User (count, active filter, avg logins)	Analytics cards
Collection averages	ElementSample (group by user, avg)	Analytics metrics
Format breakdown	ElementSample (group by format, count)	Analytics bar chart
Top collected	ElementSample (group by element, count distinct users)	Analytics ranked list
Top wanted	WishlistItem (group by element, count)	Analytics ranked list
External sources	ElementSample.source (group by source, count)	Analytics source table
Priority score	WishlistItem + WaitlistEntry + WatchlistItem (weighted formula)	Demand Signals
Waitlist growth	WaitlistEntry (weekly cohort counts)	Demand Signals line chart

8. Sidebar Navigation Discrepancy

Two **distinct sidebar navigation schemas** appear across the wireframes:

Schema A (Users-centric wireframes)

Used in: `admin-users`, `admin-users-freeze-modal`, `admin-manage-admins`, `admin-add-admin-modal`, `admin-analytics`, `admin-formats`, `admin-bulk-notification`, `admin-award-milestone`

Nav Item	Icon
Dashboard	fa-gauge
Formats	fa-shapes
Analytics	fa-chart-line
Users	fa-users
Admin Users	fa-user-shield
Export	fa-file-export

Schema B (Catalog-centric wireframes)

Used in: admin-sku-sync, admin-set-builder, admin-demand-signal

Nav Item	Icon
Overview	fa-gauge
SKU Sync	fa-arrows-rotate
Demand Signals	fa-signal
Set Builder	fa-layer-group
Inventory	fa-box
Collectors	fa-users

Resolution Needed

These are **incompatible navigation structures**. Possible interpretations:

1. **Two separate admin sections** (unlikely — no cross-linking shown)
2. **Design iteration** — Schema B is a later revision that replaces Schema A
3. **Merged navigation** — The final implementation should combine both schemas into a single sidebar

Recommendation: Merge into a unified sidebar:

```
Dashboard / Overview
Users (= Collectors)
Admin Users
Analytics
Formats
Set Builder
SKU Sync
Demand Signals
Inventory [placeholder  no wireframe]
Export
```

9. Ambiguities & Questions

Authentication & Security

#	Question	Context
Q1	What is the forgot-password flow? The link exists but no wireframe shows the reset process. Is it email-based OTP, reset link, or admin-to-admin manual reset?	Login
Q2	What session duration? How long should admin sessions last? Different for “Remember me” checked vs unchecked?	Login
Q3	Is there MFA / 2FA? No mention in wireframes, but the security note on Admin Users page suggests high sensitivity.	Manage Admins
Q4	What password policy? Min length, complexity requirements for admin accounts? The auto-generated password suggests complexity is expected.	Add Admin Modal

User Management

#	Question	Context
Q5	What does “View Details” show? The row menu has this action but no detail-view wireframe exists. What data is shown? Collection summary? Activity history?	Users
Q6	Is unfreeze a direct action or does it require a modal/reason? Freeze requires a reason; unfreeze just shows as a menu item with no confirmation step wireframed.	Users (frozen row)
Q7	Can frozen users be bulk-unfrozen? The bulk bar only shows “Freeze Accounts” — no unfreeze bulk action.	Users
Q8	What does “Export Data” export? Full user list? Filtered results? What format (CSV, Excel)? What columns?	Users
Q9	Is there a user creation flow? The admin can view users but there’s no “Add User” button. Are users only created via Shopify?	Users

Admin Management

#	Question	Context
Q10	Is “Remove” permanent deletion or soft-delete? The action exists but the consequence is unclear.	Manage Admins
Q11	Is there a confirmation modal for Deactivate/Remove? No wireframe provided.	Manage Admins
Q12	Can the primary admin be transferred? The wireframe says “You · Primary” but doesn’t show how to change primary status.	Manage Admins
Q13	Is there an admin activity/audit log? Given the security note, should admin actions be logged?	Implied

Analytics

#	Question	Context
Q14	How is “Active Users (30d)” defined? At least one login in 30 days? Or any activity (collection update, etc.)?	Analytics
Q15	Are analytics real-time or cached/computed periodically? For 247 users this is trivial, but the architecture decision matters for scale.	Analytics
Q16	What is “Wishlist” in collection metrics? The collector cabinet has Wishlist and Watchlist. Does this analytics metric combine them or is it strictly wishlist?	Analytics

Formats

#	Question	Context
Q17	What happens when deleting a format that's "Used By" users? The "Other" format has 12 users and a delete button. Does deleting cascade, reassign, or require zero users first?	Formats
Q18	How does "Add Format" work? Button exists but no wireframe shows the creation flow (modal or inline new row?).	Formats
Q19	Can format icons be customized? Each format has a FontAwesome icon, but there's no picker in the edit flow.	Formats

Set Builder

#	Question	Context
Q20	Where is the Sets list page? “← Back to Sets” breadcrumb implies a list view, but no wireframe exists.	Set Builder
Q21	How does cover image upload work? A placeholder is shown but no upload UI or file picker is wireframed.	Set Builder
Q22	What format does the set apply to? The “Transition Metals Starter Pack” example mentions “10mm cubes” in description but there’s no format selector field. Is format implicit?	Set Builder
Q23	Can sets be edited after publishing? Only “Save Draft” and “Publish Set” are shown. No “Unpublish” or “Archive” action.	Set Builder
Q24	How do sets relate to Shopify products? Does publishing a set create a Shopify product, a collection, or is it cabinet-only?	Set Builder
Q25	What are the subscription tiers? The wireframe shows “Free / Collector / Curator” but the Users page shows “Free / Lucite Pro / Curator”. Which is canonical?	Set Builder vs Users

SKU Sync

#	Question	Context
Q26	How does “Update Price” work? Does it update Shopify price, eBay price, or both? To what value?	SKU Sync
Q27	Is sync automatic or manual only? “Re-sync” button and “Last synced 12 minutes ago” suggests manual, but is there a periodic background job?	SKU Sync
Q28	What eBay API integration is needed? Are eBay credentials already configured? Which API (Browse, Inventory, Trading)?	SKU Sync
Q29	What threshold defines “Too Low” vs “Too High”? Is any negative delta “Too Low”? Or is there a percentage/dollar threshold?	SKU Sync

Demand Signals

#	Question	Context
Q30	What is the priority score algorithm? The wireframe shows scores (94, 88, 81...) but the formula is undefined. What inputs and weights?	Demand Signals
Q31	What is a “Waitlist” vs “Wishlist”? The collector cabinet has Wishlist but this page references “Waitlist” separately. Is this a new concept (notify-when-back-in-stock) or a renaming?	Demand Signals
Q32	Are demand signals actionable? No “Create Restock Order” or “Notify Waitlist” actions are shown. Are these view-only?	Demand Signals
Q33	Is “Pending Restocks” managed here or externally? The KPI shows 17 pending, but there’s no UI to mark a restock as complete or in-progress.	Demand Signals

Bulk Notifications

#	Question	Context
Q34	Can notifications include links/CTAs? The message appears text-only. Should rich content (links, buttons) be supported?	Bulk Notification
Q35	Is there a notification history/log? No wireframe shows previously sent notifications. Can admins review past broadcasts?	Implied
Q36	What email service handles “Include Email” delivery? Is this Shopify email, SendGrid, or another provider?	Bulk Notification

Milestones

#	Question	Context
Q37	Are milestones auto-awarded or always manual? The wireframe shows only manual admin-initiated awards. Should the system auto-detect “First 10 Elements” and award automatically?	Award Milestone
Q38	Can milestones be revoked? No “revoke” action is shown. What if an admin awards incorrectly?	Award Milestone
Q39	Where do milestones appear for collectors? The award is created admin-side, but no collector wireframe shows a “Milestones” or “Badges” section on the dashboard.	Award Milestone

General / Cross-cutting

#	Question	Context
Q40	What is the “Dashboard” page? Both navigation schemas list it (as “Dashboard” or “Overview”) but no wireframe exists. What KPIs/ widgets should it show?	Sidebar
Q41	What is the “Export” page? Schema A lists it in the sidebar but no wireframe exists. Is this a general data export tool?	Sidebar
Q42	What is the “Inventory” page? Schema B lists it but no wireframe exists. Is this different from SKU Sync?	Sidebar
Q43	Should the admin console be responsive? All wireframes are fixed at 1440px width. Is mobile/tablet support needed?	All screens
Q44	Is role-based access control (RBAC) needed? Currently “all admin users have full access to all features.” Will granular permissions be needed later?	Manage Admins

Appendix: File-to-Feature Matrix

Wireframe File	Primary Feature	Modals/Overlays
admin-login.html	Authentication	—
admin-users.html	User list, search, filter, bulk actions	—
admin-users-freeze-modal.html	—	Freeze Account confirmation
admin-manage-admins.html	Admin user list, activate/deactivate/remove	—
admin-add-admin-modal.html	—	Add new admin user form
admin-analytics.html	Platform metrics dashboard	—
admin-formats.html	Format CRUD with inline editing	—
admin-set-builder.html	Curated set creation with periodic table	—
admin-sku-sync.html	Cross-platform price/stock comparison	—
admin-demand-signal.html	Restock intelligence dashboard	—
admin-bulk-notification.html	—	Compose & send bulk notifications
admin-award-milestone.html	—	Award milestone badge to user